Pricing and fee structure as the player base grows

What it costs to run the game at each size, the player count at which each free limit runs out, what similar puzzle games charge, and the prices that keep the game free for most players while covering every cost.

$0
running cost today, up to about 10,000 monthly players

$15
running cost a month at 50,000 players, after free fixes

$661
running cost a month at 500,000 players

$2.99
recommended Supporter price a month ($19.99 a year)

222
supporters cover every cost at 50,000 players (under 0.5%)

1.5%
expected share who pay, based on similar games

Prepared for Bobby Taylor, owner · October 26, 2026 · Prices checked on the vendors' own pricing pages on this date (sources on the last page). Player counts are estimates from a usage model, not measurements: the live dashboard numbers should replace them once real players arrive. This is a business analysis, not tax or legal advice.

1. The short answer

> **Keep the whole game free, and add an optional Supporter tier at $2.99 a month or $19.99 a year, a one-time $4.99 Founder unlock for the first year, and $1.99–$3.99 puzzle packs.** At every size modeled, fewer than 1 player in 200 needs to pay for the game to cover all of its costs, including your build tools and help.

Why

- **Running costs stay tiny.** The servers cost nothing until about 10,000 monthly players, $5 a month until about 50,000, and about $661 a month at 500,000. That is about $1.30 per 1,000 players.
- **People and tools are the real cost.** Your design and build tools, art, help with name approvals and reports, and security are about 70–99% of the total at every size. Pricing should be set to cover those, not just servers.
- **Two costs could explode if they are not kept capped: server voices and sign-ins.** Uncapped premium voices would cost thousands a month at 100,000 players. The game already caps them and uses the free on-device voice for everyone else; keep it that way and make premium voices a Supporter perk with a monthly allowance. Firebase sign-ins start charging at about 167,000 monthly players.
- **The market sits between $2.99 and $4.99.** Puzzle Baron charges $4.99 a month, Logic Daily sells $1.99 packs and a $4.49 option, Penny Dell sells 30 puzzles for $3.99, and Everyday Grids charges $14.99 a year to remove ads. $2.99 a month is a friendly step below the leader, with no ads anywhere.

Three things to do before the game grows past 10,000 players (all free or nearly free)

1. **Move the big files off GitHub Pages.** The intro video (25 MB) and the voice engine (21 MB) will use up the Pages 100 GB a month allowance at about 10,000 players. Serving them from the Cloudflare Worker's static files is free with no limit. GitHub's rules also say Pages is not for running an online business, so once money is involved, the whole site is safer on Cloudflare Pages (free, commercial use allowed, works with your domain without moving it to Cloudflare).
2. **Move player emails off Gmail before the daily email passes about 500 a day.** Amazon SES costs about $0.10 per 1,000 emails. costs money
3. **Switch Cloudflare to the $5 Workers Paid plan at about 10,000 monthly players.** One plan covers the Worker, the database and higher limits. costs money The admin dashboard's free-limit alerts already warn at 70%.

Everything marked costs money needs your OK first, under the free-only rule. Nothing in this report has been bought or switched on.

2. What the game uses today, and what is free

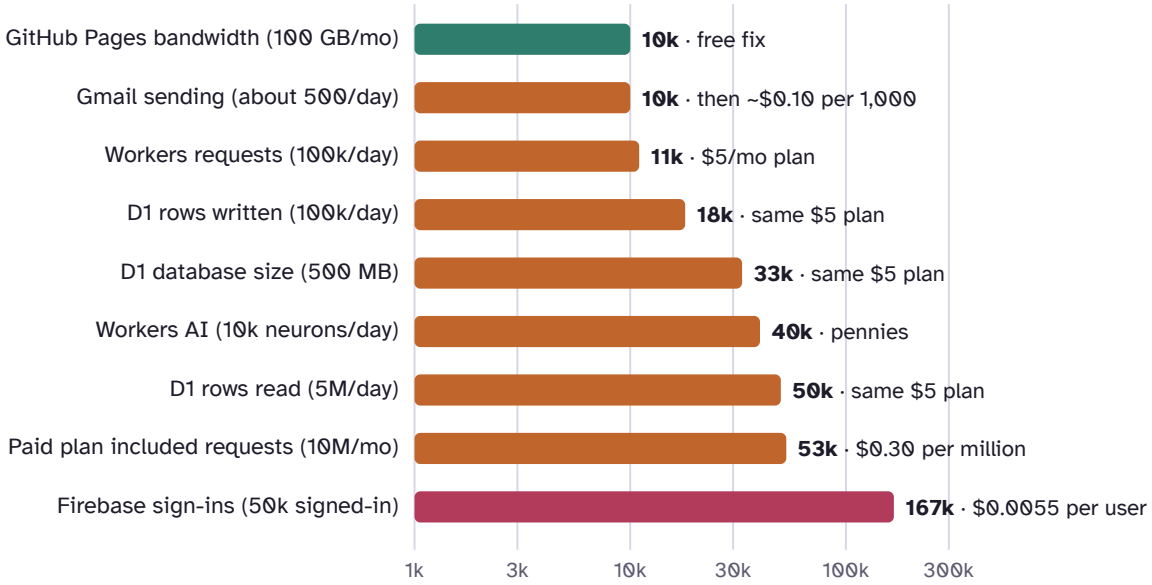
Every service below is on a free tier now. The game already has caps on the services that bill past their free amount.

Service	What it does	Free amount	Price after the free amount
Cloudflare Workers	The game's server (sign-in checks, saves, races, daily puzzle, admin)	100,000 requests a day	$5 a month plan: 10 million requests and 30 million CPU-ms a month included, then $0.30 per million requests
Cloudflare D1	The database	5 million rows read and 100,000 written a day; 500 MB per database, 5 GB in all	On the $5 plan: 25 billion reads and 50 million writes a month included, then $1 per million writes; 5 GB included, then $0.75 per GB
Cloudflare Workers AI	Story pictures, translation, rating summaries	10,000 "neurons" a day	$0.011 per 1,000 neurons. Pictures are cached, so each one is made once.
Cloudflare static files and R2	Not used for the game yet; recommended for big files	Static files: free and unlimited. R2: 10 GB stored, no download fees	R2: $0.015 per GB a month
Firebase Authentication	Optional player accounts	50,000 monthly signed-in players	$0.0055 each up to 100,000, then $0.0046 up to 1 million
Gemini voices (Google AI Studio)	Natural character voices, first choice	Free tier with daily request limits; the game caps new clips at 300 a day	Paid tier about $9 per million audio tokens (about $0.81 an hour of speech) until December 31, 2026; it doubles on January 1, 2027
Google Cloud Chirp 3 HD	Backup natural voice	1 million characters a month; the game stops at 900,000	$30 per million characters
Kokoro (on the player's device)	Natural voice when server voices are capped	Free, runs in the browser	None. Costs only download bandwidth.
GitHub Pages	The website at thelogicgrid.com	100 GB a month download (soft limit), 1 GB site	No paid tier; you move to another host. Not meant for online businesses.
GitHub Actions	Builds, nightly checks, reports, owner emails	Unlimited on a public repository	Private repository: 2,000 minutes a month free, then about $0.006 a minute (prices cut about 40% in January 2026)
Gmail (sending)	Archive, reports, player emails	About 500 recipients a day	Move to Amazon SES ($0.10 per 1,000) or Resend ($20 a month for 50,000)
Domain thelogicgrid.com	The web address	None	About $12–$15 a year at renewal

The game does not use the paid parts of any service. The admin dashboard tracks Cloudflare and Firebase use against these limits (server/src/usage.js), and the nightly run emails you when something needs attention.

3. When each free limit runs out

The chart shows the number of **monthly active players** at which each free limit is reached. The model assumes a quarter of monthly players play on a given day (normal for a daily-puzzle game), 25 server requests and 15 database writes per daily player, about 10 MB of website download per monthly player, and 30% of players signing in. Busy days are counted at 1.5 times a normal day, so the limits are reached a little before the averages say.

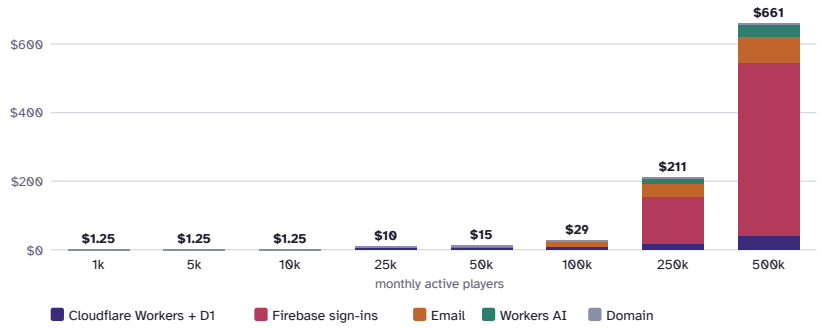


Green: a free fix exists. Orange: covered by one $5 a month plan or pennies. Red: starts billing per player. Log scale.

What happens at each limit

- **About 10,000 players: website bandwidth and Gmail.** Both have a free or nearly free fix (see section 1). These come first because the video and voice engine are large downloads.
- **11,000–50,000 players: Cloudflare's daily limits.** Requests, database writes, database size and reads all move to the same $5 a month Workers Paid plan. Without it, the server starts refusing requests on busy days.
- **About 40,000 players: Workers AI.** New pictures and translations start to cost cents a month.
- **About 53,000 players: the $5 plan's included requests.** Each extra million requests costs $0.30, which is about $1.90 for every extra 10,000 players.
- **About 167,000 players: Firebase sign-ins.** This is the first cost that grows by the player: about $137 a month at 250,000 players and $505 at 500,000. A sign-in system built into the Worker (email links or passkeys) would make this free again; it is worth building at around 100,000 players.
- **Server voices never run out, because they are capped.** When the caps are reached, players hear the free on-device voice instead.

4. Monthly cost at each stage of growth



Running costs only (servers, sign-ins, email, domain), after the free fixes in section 1.

Monthly players	1k	5k	10k	25k	50k	100k	250k	500k
Daily players (25%)	250	1,250	2,500	6,250	12,500	25,000	62,500	125,000
Signed-in players (30%)	300	1,500	3,000	7,500	15,000	30,000	75,000	150,000
Running costs a month								
Cloudflare Workers + D1	$0	$0	$0	$5.00	$5.00	$7.78	$17	$41
Workers AI (pictures, translation)	$0	$0	$0	$0	$0.82	$4.95	$17	$38
Firebase sign-ins	$0	$0	$0	$0	$0	$0	$138	$505
Email (Amazon SES after Gmail)	$0	$0	$0	$3.75	$7.50	$15	$38	$75
Website hosting (after free fix)	$0	$0	$0	$0	$0	$0	$0	$0
Server voices (kept at free caps)	$0	$0	$0	$0	$0	$0	$0	$0
Domain (about $15 a year)	$1.25	$1.25	$1.25	$1.25	$1.25	$1.25	$1.25	$1.25
Running costs total	**$1.25**	**$1.25**	**$1.25**	**$10**	**$15**	**$29**	**$211**	**$661**
People, tools and safety a month (estimates you control)								
Design and build tools (AI coding plan)	$100	$100	$100	$100	$200	$200	$200	$200
Art, music and voice extras	$0	$0	$100	$100	$100	$300	$300	$300
Help with names, reports, support	$0	$0	$0	$100	$100	$500	$1,000	$1,000
Security review and insurance	$0	$0	$0	$0	$50	$50	$150	$150
Total a month	**$101**	**$101**	**$201**	**$310**	**$465**	**$1,079**	**$1,861**	**$2,311**
Cost per 1,000 players	$101.25	$20.25	$20.12	$12.40	$9.29	$10.79	$7.44	$4.62
Supporters needed to break even	49	49	96	148	222	514	887	1,101
...as a share of players	4.8%	0.96%	0.96%	0.59%	0.44%	0.51%	0.35%	0.22%

People, tools and safety lines are planning estimates, not bills. "Design and build tools" assumes an AI coding plan at $100 a month, rising to $200 for heavier use. "Help" is part-time paid help with screen-name approvals, player reports and support, which grows with players because typed names wait for admin approval. "Security" covers an outside review and small-business cyber insurance once money is taken. Supporters needed assumes $2.10 left over a month per paying supporter after fees (section 6).

5. The costs to keep capped

Server voices

The natural server voices are the one feature that could cost real money fast. The table shows the monthly bill if the caps were removed and 1 in 10 daily players heard a new half puzzle read aloud each day (about 750 characters or 50 seconds of speech).

Monthly players	10k	25k	50k	100k	250k	500k
Chirp 3 HD, uncapped	$139	$392	$814	$1,658	$4,189	$8,408
Gemini Flash TTS, uncapped (2026 price)	$84	$211	$422	$844	$2,109	$4,219
Gemini Flash TTS, uncapped (from Jan 2027)	$168	$422	$844	$1,688	$4,218	$8,438
As built today (caps on, Kokoro on the device)	**$0**	**$0**	**$0**	**$0**	**$0**	**$0**

> **Keep the caps on.** Today the game spends $0 on voices: it uses Gemini's free tier up to 300 new clips a day, Chirp up to 900,000 characters a month, and the free Kokoro voice on the player's device after that. Clips are saved and reused, so the daily puzzle is voiced once for everyone. If you want more premium voice time, give it to Supporters with a monthly allowance, and raise the caps only by what supporter income pays for. Gemini's price doubles on January 1, 2027, so revisit then.

Other costs that grow with success

- **Sign-ins past 167,000 players** (section 3). Plan a free Worker-based sign-in around 100,000 players.
- **Making the repository private** (being weighed in "Protect the game from copying"): GitHub Pages from a private repository needs GitHub Pro at $4 a month, and Actions get 2,000 free minutes a month. The nightly checks, video builds and reports can use more than that, about $5–$20 a month extra. Moving the website to Cloudflare Pages avoids the $4. costs money
- **Code scanning** is free on a public repository. On a private one, GitHub's code scanning costs $30 per person a month; free tools such as Semgrep Community can replace it.
- **Mobile apps** (paused): Apple charges $99 a year and takes 15% of sales for small businesses; Google Play is $25 once and 15%. The Microsoft Store is free for individual developers.
- **Moderation time** grows with players because typed screen names wait for approval. Suggested names auto-approve, so steering players toward them keeps this cost down.

6. What players pay for similar games

Game	Free part	Paid	Model
Puzzle Baron's Logic Puzzles	About 100 free puzzles	$2.99 volumes; $4.99 a month	Packs plus subscription
Logic Daily	Daily puzzle	$1.99 packs; $4.49 option	Packs
Penny Dell Logic Puzzles	A few samples	$3.99 for 30 puzzles	Packs
Everyday Grids	Everything, with ads	$14.99 a year to remove ads	Ads plus yearly
The Fantastical Logic Grid (now)	Everything, endless new puzzles, no ads	Nothing yet	Free

Competitor prices from the App Store on October 26, 2026 (market test).

What that means

- Players of logic puzzles are used to paying **$2–$5 for a pack** and **about $5 a month** at the top end. A yearly price of $15–$20 is normal.
- This game's edge is that every puzzle is new and nothing is locked. Locking puzzles behind a paywall would throw that away. Selling **extras and thanks** keeps it.
- For free games with an optional paid tier, 1–3% of monthly players usually pay; tip jars alone usually get under 0.5%. This report uses 0.5%, 1.5% and 3%.

What you keep from each sale

Sale	Lemon Squeezy (handles sales tax)	Ko-fi free plan	GitHub Sponsors
$2.99 a month	$2.33	$2.45	$2.99
$19.99 a year	$18.39 ($1.53 a month)	$18.11	$19.99
$4.99 one time	$4.24	$4.30	$4.99
$1.99 pack	$1.39	$1.53	n/a (no shop)

Lemon Squeezy: 5% + 50¢, plus 0.5% on subscriptions and 1.5% for buyers outside the US. Ko-fi: 5% on memberships and shop on the free plan (0% on Ko-fi Gold at $12 a month), plus Stripe or PayPal at about 2.9% + 30¢. GitHub Sponsors: no fee on personal accounts. A blend of monthly, yearly and one-time supporters leaves about $2.10 a month per supporter, which the break-even numbers use.

7. Recommended fee structure

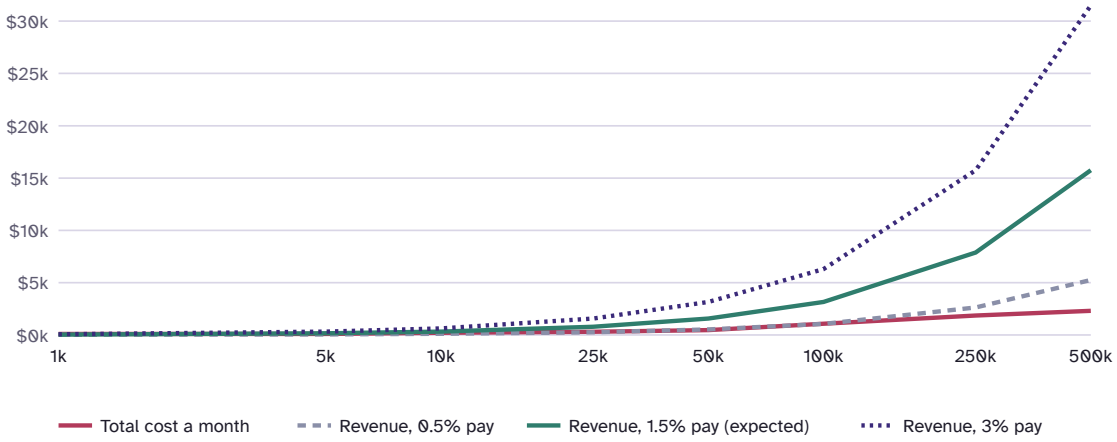
Tier	Price	What it includes	Why this price
Free	**$0, forever**	Every puzzle, theme, size and difficulty; daily puzzle and streaks; races and scoreboard; hints, saving, printing, notes; on-device voices; accounts and sync; every language and accessibility feature. No ads.	The free game is the marketing. It is what makes people share it.
Tip jar	**Any amount ($3 suggested)**	A thank-you and, if signed in, the Supporter badge for a month.	Costs nothing to run. GitHub Sponsors takes no fee.
Supporter	**$2.99 a month or $19.99 a year**	Supporter badge on the scoreboard; a monthly allowance of premium server voices; early access to new themes; longer stats history and extra badges; a monthly printable puzzle book.	A step under Puzzle Baron's $4.99, and the yearly price sits with Everyday Grids. The voice allowance is a perk that costs little because it is capped.
Founder (first 12 months only)	**$4.99 one time**	Lifetime Supporter badge and all bonus packs released in year one. Premium voices not included beyond a small monthly allowance.	Rewards early fans without a lifetime promise of costs that grow.
Puzzle packs and books	**$1.99 (25 puzzles) or $3.99 (60-puzzle printable book)**	Hand-picked themed sets with answer keys, as print-ready PDFs.	Matches Logic Daily's $1.99 and Penny Dell's $3.99. Costs nothing to deliver.

Rules that protect players and the budget

- Never lock a core feature or puzzle. Paid things are extras, thanks, and convenience.
- No ads before 50,000 monthly players, and only if supporter income falls short. Ads earn about $1–$5 per 1,000 visits and hurt a calm puzzle game.
- Use a seller that handles sales tax for you (Lemon Squeezy or Paddle) for anything sold. Tips can go through GitHub Sponsors.
- Check prices once a year and when a vendor changes prices (next: Gemini voices on January 1, 2027).
- Keep one switch for each paid extra in the admin Settings, as with the Support link, so anything can be turned off without a release.

> **How this fits the work in progress.** The "Prepare freemium launch package" work is building three money options for you to choose from: a tip jar with a Supporter badge, a Ko-fi membership, and a Lemon Squeezy one-time unlock. This report recommends starting with the tip jar and Supporter badge (no cost, no code beyond the badge), then adding the $2.99 / $19.99 Supporter tier and the $4.99 Founder unlock through Lemon Squeezy once there are about 2,000 monthly players to sell to. The Support the game switch from the freemium launch is already in the admin dashboard.

8. Will it pay for itself?



Revenue at three shares of paying players (at $2.10 net each a month) against the total monthly cost from section 4. Players on a log scale; the dollar scale is cut at $33k.

Monthly players	1k	5k	10k	25k	50k	100k	250k	500k
Revenue if 0.5% pay	$10	$52	$105	$262	$525	$1,050	$2,625	$5,250
left after all costs	−$91	−$49	−$96	−$48	+$60	−$29	+$764	+$2,939
Revenue if 1.5% pay	$32	$158	$315	$788	$1,575	$3,150	$7,875	$15,750
left after all costs	−$70	+$56	+$114	+$478	+$1,110	+$2,071	+$6,014	+$13,439
Revenue if 3% pay	$63	$315	$630	$1,575	$3,150	$6,300	$15,750	$31,500
left after all costs	−$38	+$214	+$429	+$1,265	+$2,685	+$5,221	+$13,889	+$29,189

- **Below about 3,000 players** supporters will not cover your build tools, even at 3%. That is normal for a new game, and running costs are $0, so nothing is lost while it grows.
- **From about 3,200 players** the expected case (1.5%) covers every cost in the model, and from about 25,000 players even the low case (0.5%) does, including paid help and security.
- **At 50,000 players** only 222 supporters are needed; at 500,000, 1,101 (about 0.2%).

9. Plan by stage

Stage	Do this	New monthly cost
Now to 2,000 players	Launch free. Turn on the Support the game link with a tip jar (GitHub Sponsors). Move the big files to Cloudflare. Watch the free-limit alerts.	$0 free
2,000 to 10,000	Add the Supporter tier and Founder unlock through Lemon Squeezy. Release the first puzzle book. Move player email to Amazon SES.	About $1 costs money
10,000 to 50,000	Switch to Cloudflare Workers Paid. Move the website to Cloudflare Pages. Budget for occasional art and part-time help with name approvals.	$5 to $15 running costs money
50,000 to 150,000	Build a free sign-in in the Worker to avoid Firebase charges. Add a security review and cyber insurance. Consider premium voice allowance increases paid by supporters.	$15 to $50 running
150,000 and up	Review prices. Consider the mobile apps (resume only on your word). Consider a small team for support.	$200 to $700 running

10. Assumptions and sources

Usage model

Daily players	25% of monthly players	Server requests	25 per daily player a day (config, saves, daily puzzle, scores, events), 2 ms CPU each
Database writes	15 per daily player a day	Database reads	400 per daily player a day
Database size	15 KB per monthly player, kept small by the weekly archive	Busy-day factor	1.5 times an average day
Signed in	30% of monthly players	Website download	10 MB per monthly player a month (first visits, updates, video and voice engine)
Daily email	5% of monthly players subscribe	Workers AI	1 neuron per daily player a day (pictures are cached)
Paying supporters	0.5%, 1.5% or 3% of monthly players	Net per supporter	$2.10 a month after fees, a blend of monthly, yearly and one-time

Replace these with real numbers after the first month: the admin dashboard shows requests, database rows, Firebase users and email counts. The model's source is kept with this report so it can be re-run.

Sources (checked October 26, 2026)

- Cloudflare Workers pricing: developers.cloudflare.com/workers/platform/pricing/ ($5 minimum, 10M requests and 30M CPU-ms included, static assets free and unlimited)
- Cloudflare D1 pricing: developers.cloudflare.com/d1/platform/pricing/
- Cloudflare R2, Workers AI and KV pricing: cloudflare.com/pricing/
- Firebase Authentication tiers: firebase.google.com/pricing (as summarized by rapidnative.com, 2026)
- Google Cloud Text-to-Speech pricing (Chirp 3 HD $30 per million characters, 1 million free): cloud.google.com/text-to-speech/pricing
- Gemini Flash TTS pricing and the January 2027 increase: ai.google.dev/pricing (as summarized by eesel.ai, 2026)
- GitHub Pages limits and commercial use: docs.github.com/en/pages/getting-started-with-github-pages/github-pages-limits
- GitHub Actions 2026 price changes: heise.de (December 2025)
- Gmail sending limits: support.google.com/mail/answer/22839
- Resend pricing: resend.com/pricing · Amazon SES pricing: aws.amazon.com/ses/pricing
- Lemon Squeezy fees: lemonsqueezy.com/pricing (as summarized by swell.is) · Ko-fi Gold $12 a month: ko-fi.com (as summarized by schoolmaker.com)
- Competitor prices: App Store listings, October 26, 2026 (project market test)
- Game limits and caps: server/src/usage.js (FREE_LIMITS), server/wrangler.toml (DAILY_TTS_CAP, GCLOUD_TTS_MONTH_CAP), docs/launch/FREEMIUM.md